Econ 182: Presenting Early Results for Market Tightness' Effect on Incarceration Rates in the United States

Matthew Ramos

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Review the Research Questioning and Project Positioning

- The goal of this research is to analyze the effects that Labor Market Tightness has on incarceration rates
  - We will look at the incarceration levels (per 100k people) and the LMT from 2001 to 2022
  - Concerns public policy makers and those in the workforce during harsher periods in the business cycle

Presentation of Research Methodology

- First, I have calculated the tightness of the labor market based on job openings and unemployment data from the Bureau of Labor Statistics (JOLTS)
  - I have also found the rates of incarceration at the federal level per 100k individuals
  - Regress Incarceration with LMT to see the effect that LMT has on a statistical level
  - Develop a simple regression equation to evaluate the variables of interest

$$\text{Inc} = \beta_0 + \beta_1 \frac{V}{U} + \beta_n \Omega + E(0) \tag{1}$$

Presentation of Raw Data

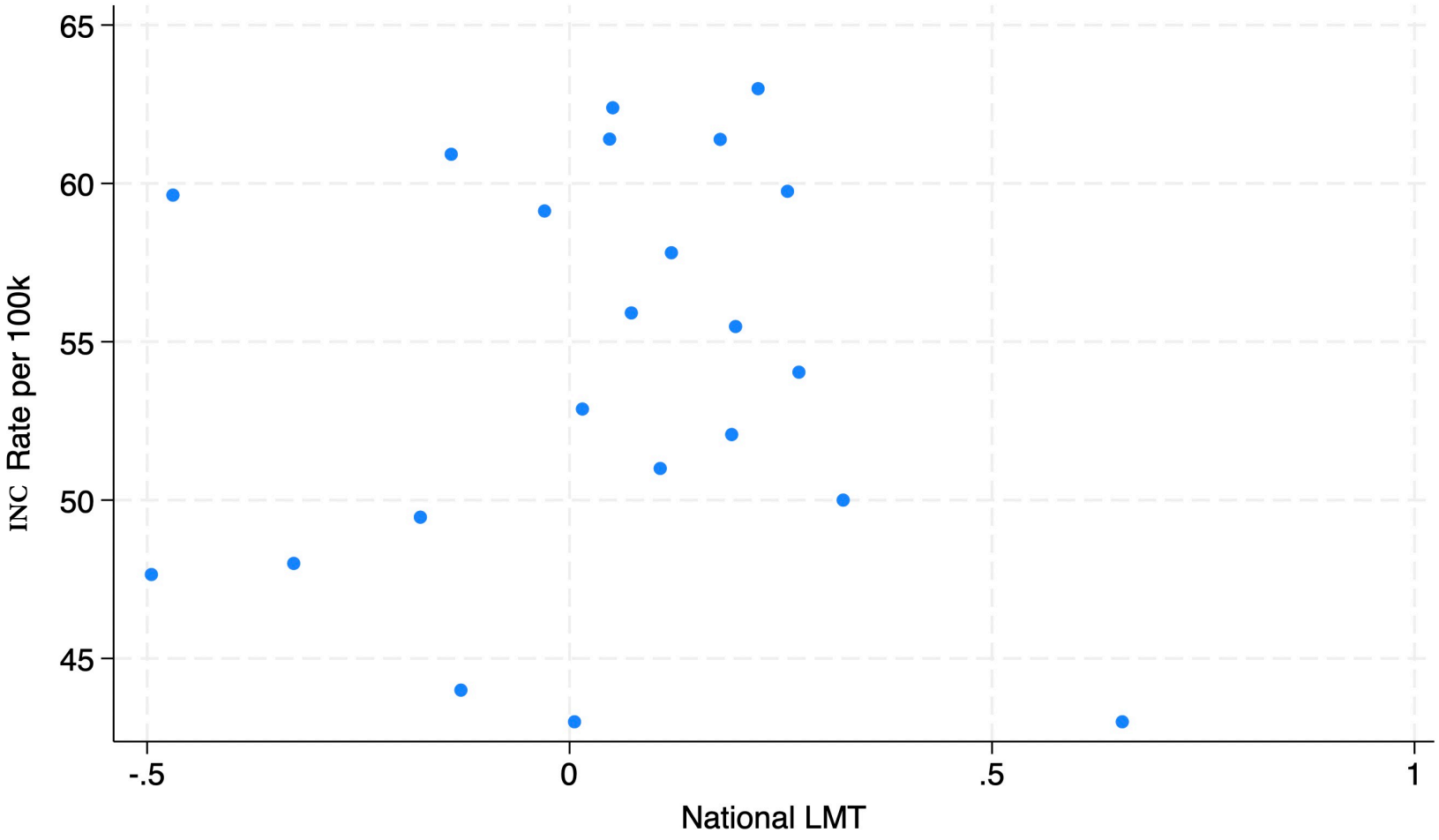

Figure: Visualized Raw Data

Presentation of Data: Time Series

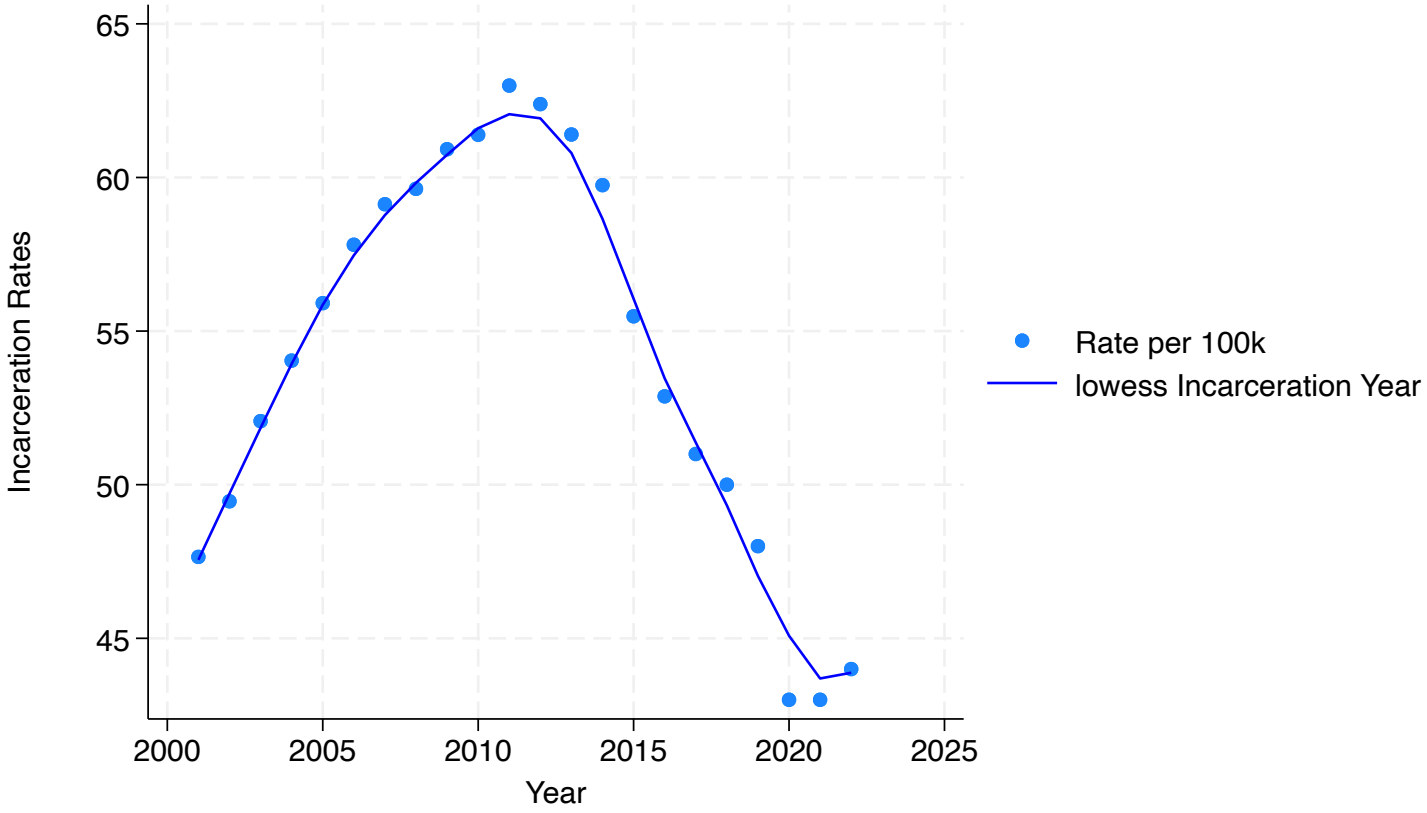

Figure: Federal Incarcerations

Presentation of Raw Data: Time Series

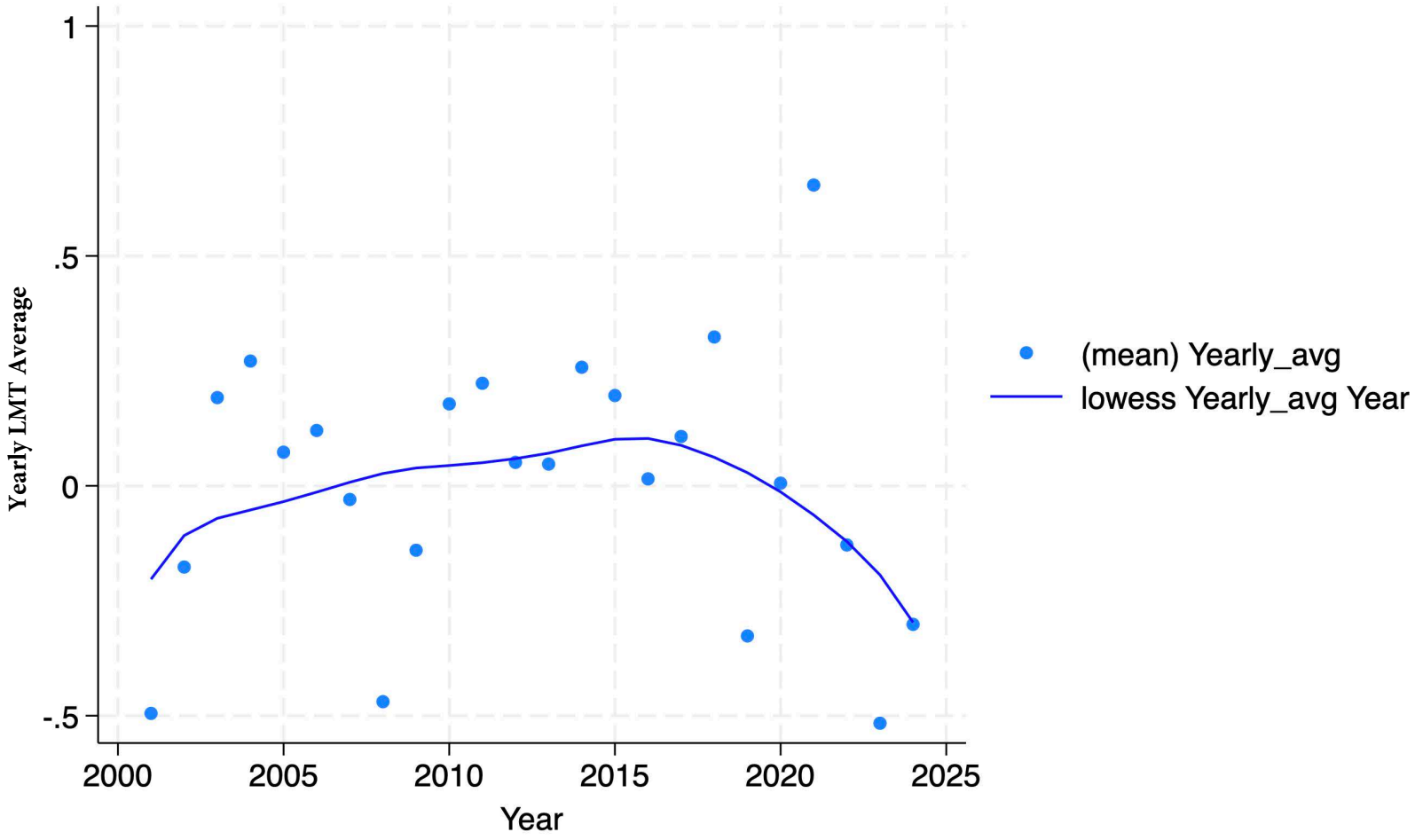

Figure: Average LMT

Presentation of Early Results: Numbers

Column1	(1)
VARIABLES	Incarceration Regression
Yearly_avg	3.119
	(0.215)
Inc_State	0.215***
	(0.000)
Year	0.762***
	(0.000)
Constant	-1,566.196***
	(0.000)
Observations	22
R-squared	0.840
pval in parentheses	
*** p<0.01, ** p<0.05, * p<0.1	

Figure: Early Estimation

Conclusion: Hurdles + Steps

- Hurdle 1:
  - Formatting in Stata has been troublesome, especially with multiple datasets
- Hurdle 2:
  - finding the best range of data bin widths to get the most informative and accurate results and graphs
- Next Steps:
  - Produce the same results at the state level for a few states to try to find similar or more distinct patterns
  - Introduce more of the data variables that could help reduce noise and bias